E-COMMERCE SALES ANALYSIS

Business Insights & Recommendations

End-to-End Exploratory Data Analysis

Prepared from the cleaned e-commerce sales dataset

100	21	$76,931.28	275
Records	Columns	Total Sales	Units Sold

1. Project Overview

This project analyses an e-commerce sales dataset to understand overall sales performance, product and category contribution, customer locations, payment methods, order status, customer segments and monthly sales trends. The objective is to convert cleaned transaction data into clear business insights that can support better sales, marketing and operational decisions.

2. Dataset Overview

Item	Details
Records	100
Columns	21
Period	January–April 2024
Total Sales	$76,931.28
Quantity Sold	275 units
Average Sales per Order	$769.31

3. Data Cleaning & Preparation

Before performing exploratory analysis, the cleaned dataset was checked to ensure that the fields required for analysis were available and usable.

- Checked the dataset structure, column names and data types.
- Checked missing values across the dataset.

- Checked for duplicate records.
- Validated key numeric fields including quantity, unit price, discount, shipping cost, profit margin and total sales.
- Used the date and month fields for time-based analysis.
- Used the cleaned dataset consistently for all subsequent exploratory analysis.

4. Exploratory Data Analysis Performed

- Overall sales and quantity analysis.
- Sales performance by product category.
- Sales performance by individual product.
- Sales by customer location.
- Sales by payment method.
- Sales by customer segment and gender.
- Monthly sales trend.
- Order-status analysis.
- Review-rating and profit-margin analysis.

5. Key Business Insights

1. **Sports is the highest-revenue category.** Sports generated $19,059.49, the largest category contribution in the dataset, followed by Electronics at $15,271.19 and Beauty at $14,609.57.
2. **Smartphones are the strongest individual product by sales.** Smartphones generated $17,519.83, followed by T-Shirts at $16,844.58 and Lipstick at $15,768.89.
3. **New York is the leading customer location.** New York generated $18,271.37 in sales, followed by Los Angeles at $17,542.30 and Houston at $15,308.11.
4. **Debit Card is the leading payment method by sales.** Debit Card transactions contributed $23,813.36, ahead of Cash on Delivery at $18,604.46.
5. **Sales are concentrated in the first three months.** March generated the highest monthly sales at $23,657.05, followed by January at $23,031.80 and February at $22,475.11. April was substantially lower at $7,767.32.
6. **Order-status performance requires attention.** The dataset contains 40 cancelled orders and 33 returned orders compared with 27 completed orders, suggesting that order completion and retention should be investigated further.

6. Data-Driven Recommendations

1. **Prioritise high-performing categories and products.** Maintain strong stock availability and targeted promotions for Sports, Smartphones and other high-revenue products.
2. **Strengthen regional marketing.** Use the strong sales contribution from New York, Los Angeles and Houston to plan location-specific campaigns and inventory allocation.
3. **Investigate cancellations and returns.** Review the reasons behind cancelled and returned orders and identify operational or product issues that can improve completed-order rates.
4. **Optimise payment options.** Continue supporting Debit Card while monitoring other payment methods to understand customer preferences and reduce payment friction.

5. **Address the April sales decline.** Compare April's product, category and order-status performance with earlier months and use targeted offers or campaigns to improve weaker periods.

7. Conclusion

The analysis shows clear differences in performance across categories, products, locations and payment methods. Sports and Smartphones are important revenue contributors, while New York is the strongest customer location. The sharp reduction in April sales and the high number of cancelled and returned orders are the most important areas for further business investigation. These findings can help management focus inventory, marketing and operational improvements on the areas with the greatest potential impact.

8. Acitivity Summary

This project demonstrates an end-to-end analytical workflow: preparing and validating a cleaned sales dataset, performing exploratory data analysis in Python, identifying business trends, and translating findings into actionable recommendations. The analysis focuses on practical business questions rather than only reporting raw numbers.

Submission note: This report is intentionally focused on data cleaning, exploratory analysis, business insights and recommendations.